Digital Thermometer with LCD

Project Overview

This project implements a Digital Thermometer using an Arduino Uno, TMP36 Temperature Sensor, and a 16×2 LCD Display. The system continuously measures ambient temperature and displays the temperature value in degrees Celsius on the LCD screen.

The project was designed and tested using Tinkercad and demonstrates sensor interfacing, analog-to-digital conversion, and LCD communication using Arduino.

Project Details

Project Title: Digital Thermometer with LCD

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Components Required

- Arduino Uno
- TMP36 Temperature Sensor
- LCD 16×2 Display
- 10kΩ Potentiometer
- Breadboard
- Jumper Wires
- USB Cable

Circuit Connections

TMP36 Sensor

TMP36 Pin	Arduino
VCC	5V
OUT	A0

TMP36 Pin	Arduino
GND	GND

LCD Display

LCD Pin	Arduino Pin
RS	12
EN	11
D4	5
D5	4
D6	3
D7	2
VSS	GND
VDD	5V
RW	GND

Potentiometer

- Left Pin → GND
- Middle Pin → LCD V0
- Right Pin → 5V

Working Principle

The TMP36 sensor senses the surrounding temperature and generates an analog voltage proportional to the temperature.

The Arduino Uno reads this voltage through analog pin A0 and converts it into a temperature value using the formula:

Temperature (°C) = (Voltage − 0.5) × 100

The calculated temperature is displayed on the 16×2 LCD screen and updates continuously in real time.

Arduino Code

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(12,11,5,4,3,2);

int sensorPin = A0;

void setup()
{
  lcd.begin(16,2);
}

void loop()
{
  int reading = analogRead(sensorPin);

  float voltage = reading * 5.0;
  voltage /= 1024.0;

  float temperatureC = (voltage - 0.5) * 100;

  lcd.clear();

  lcd.setCursor(0,0);
  lcd.print("Temperature");

  lcd.setCursor(0,1);
  lcd.print(temperatureC);
  lcd.print((char)223);
  lcd.print("C");

  delay(1000);
}
```

Sample Output

```
Temperature
24.71°C
```

Applications

- Room Temperature Monitoring
- Weather Monitoring Systems
- Industrial Automation
- Medical Equipment
- HVAC Systems
- Smart Home Projects

Advantages

- Low Cost
- Easy Implementation
- Real-Time Temperature Monitoring
- Accurate Measurement
- Educational and Practical Learning Project

Future Improvements

- IoT-Based Temperature Monitoring
- Cloud Data Storage
- Mobile Application Integration
- Temperature Alert System
- Wireless Communication Module

Tools Used

- Tinkercad Circuit Simulator
- Arduino IDE
- GitHub

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